

08.1

A scientist measured the pressure in a phloem tube in a willow plant stem. He repeated his measurements to obtain nine readings.

His results are shown in **Table 3**.

Table 3

Phloem pressure / arbitrary units								
7.4	8.0	7.0	8.6	8.2	9.3	7.4	9.1	8.8

The percentage error of the mean phloem pressure in this phloem tube is calculated using this equation.

$$\text{Percentage error} = \frac{\text{uncertainty in measurement}}{\text{mean}} \times 100$$

The uncertainty in measurement is half the range of the measured values.

Calculate the percentage error of the mean phloem pressure in this phloem tube.

Show your working.

[2 marks]

Percentage error _____ %



0	8	.	2
---	---	---	---

The mass flow hypothesis is used to explain the movement of substances through phloem.

Use your understanding of the mass flow hypothesis to explain how pressure is generated inside this phloem tube.

[3 marks]

Question 8 continues on the next page

Turn over ►

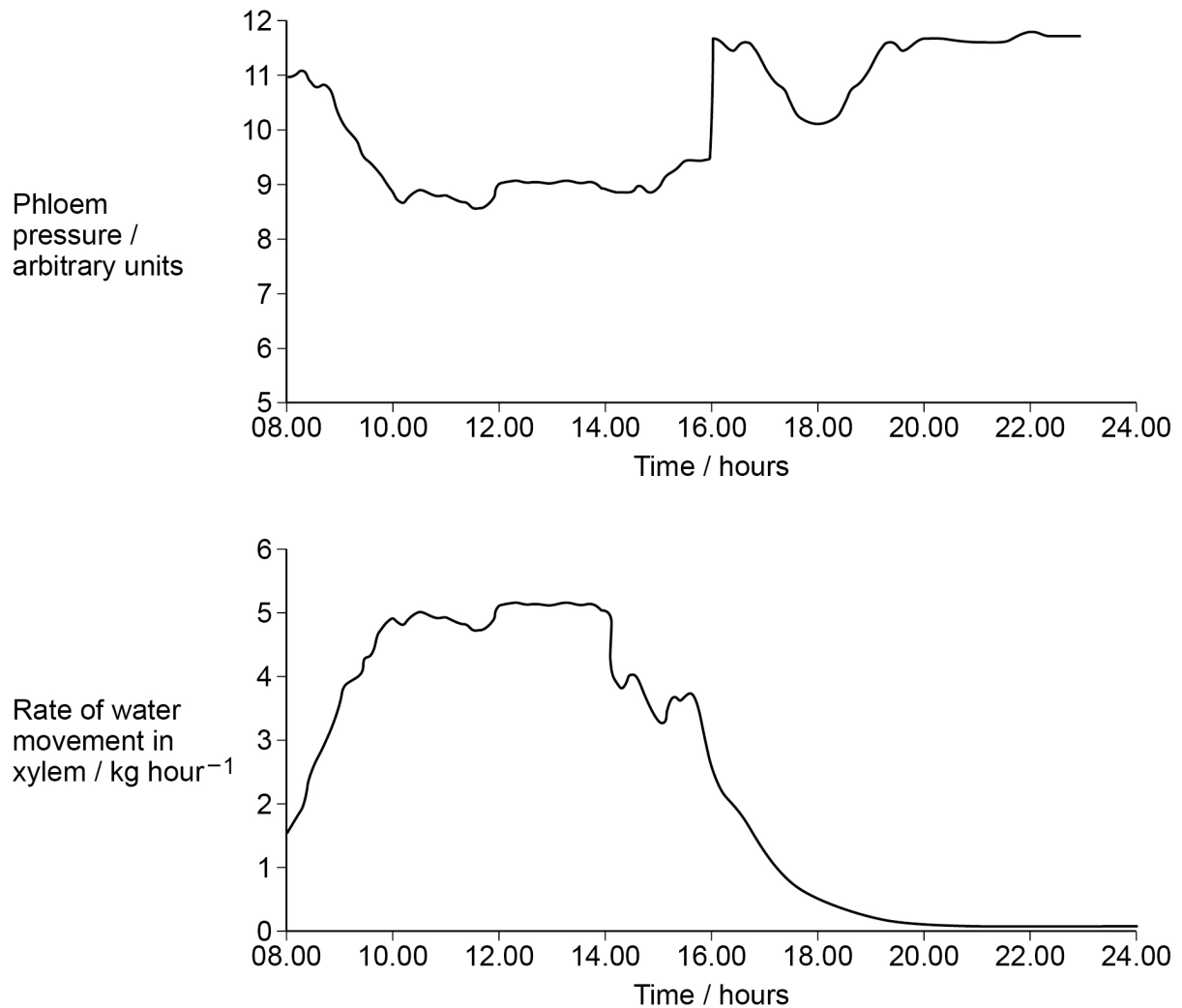


08.3

The scientist also measured changes in the phloem pressure and changes in the rate of water movement in the xylem of a willow plant at intervals during a day.

His results are shown in **Figure 6**.

Figure 6



Describe the relationship between phloem pressure and the rate of water movement in xylem in this plant.

[1 mark]



0 8 . 4

Phloem pressure is reduced during the hottest part of the day. Use information in **Figure 6** along with your understanding of transpiration and mass flow to explain why.

[3 marks]

9

Turn over for the next question

Turn over ►